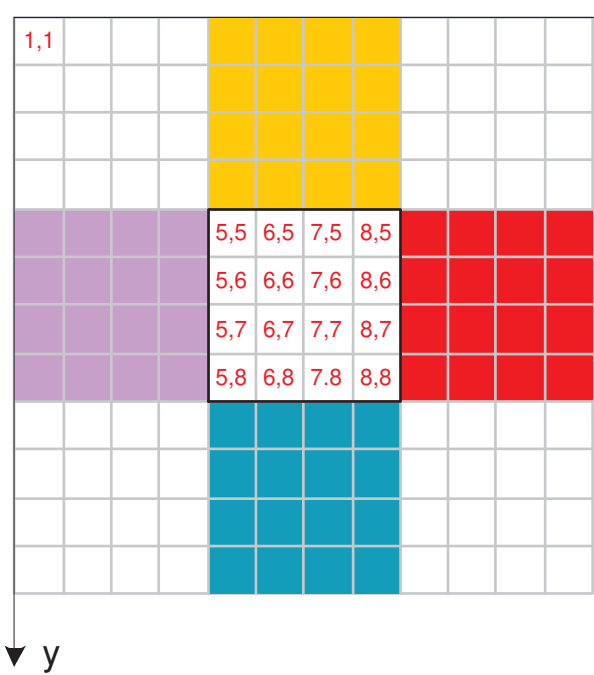
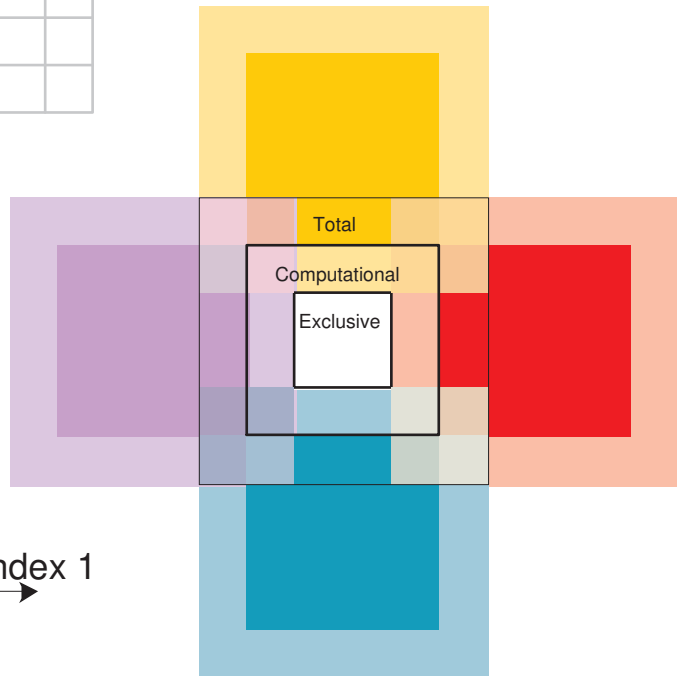




This example shows a grid with a length of 12 cells along the x-axis and a length of 12 cells along the y-axis. The grid is decomposed over a DELayout that has a length of 3 DEs along the x-axis and 3 DEs along the y-axis. We will look at how halo domains are defined for the data assigned to the central DE. The indices shown in red are global grid indices in the form (x,y).

We will assume that the data on each DE depends on a nearest neighbor in each direction (N,S,E,W). In order to perform computations efficiently, we would like this data on the local DE. To do this we specify a halo width of 1 cell in all directions for the data on each DE at Array or Field creation. Extra memory is allocated to hold the replicated grid cells.



Axis Index 1

1,1	2,1	3,1	4,1	5,1	6,1
1,2	2,2	3,2	4,2	5,2	6,2
1,3	2,3	3,3	4,3	5,3	6,3
1,4	2,4	3,4	4,4	5,4	6,4
1,5	2,5	3,5	4,5	5,5	6,5
1,6	2,6	3,6	4,6	5,6	6,6

Axis Index 2

The diagram directly left shows index values for the data on the central DE. The global grid indices are shown in red. Axis indices, which correspond to the memory allocated on the DE, are shown in blue. The **exclusive** domain is the inner black square; it contains grid cells that are not replicated on any other DEs. The black square outlined darkly in black is the **computational** domain; it contains all of the grid cells that have unique global grid indices and are updated by the local DE. The **total** domain is the entire extent of the memory allocated on the local DE.